

Homework 1

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Problem 1

$$(a) 2u - v = [4, -2, 6]^T - [1, 4, -2]^T = [3, -6, 8]^T$$

$$(2u - v) \cdot w = 3(-4) + (-6)(2) + 8(6) = -12 - 12 + 48 = 24$$

Ans: 24

$$(b) (1/2)w = [-2, 1, 3]^T \rightarrow v + (1/2)w = [1, 4, -2]^T + [-2, 1, 3]^T = [-1, 5, 1]^T$$

$$||[-1, 5, 1]^T|| = \sqrt{1 + 25 + 1} = \sqrt{27} = 3\sqrt{3} \approx 5.20$$

Ans: $3\sqrt{3} \approx 5.20$

$$(c) \cos \theta = (u \cdot w) / (||u|| ||w||)$$

$$u \cdot w = 2(-4) + (-1)(2) + 3(6) = -8 - 2 + 18 = 8$$

$$||u|| = \sqrt{4 + 1 + 9} = \sqrt{14} \rightarrow ||w|| = \sqrt{16 + 4 + 36} = \sqrt{56} = 2\sqrt{14}$$

$$\cos \theta = 8 / (\sqrt{14} \cdot 2\sqrt{14}) = 8/28 = 2/7 \rightarrow \theta = \arccos(2/7) = 73.398\dots^\circ$$

Ans: $\theta \approx 73.40^\circ$

Problem 2

(a) AB undefined: A is 3×3 so it has 3 cols, B is 2×3 so it has 2 rows, $3 \neq 2$.

$$BA \text{ is } 2 \times 3 \rightarrow r1 = [1(2) + 0(1) - 2(0), 1(-1) + 0(3) - 2(4), 1(0) + 0(-2) - 2(1)] = [2, -9, -2]$$

$$r2 = [3(2) + 1(1) + 1(0), 3(-1) + 1(3) + 1(4), 3(0) + 1(-2) + 1(1)] = [7, 4, -1]$$

$$\text{Ans: } AB \text{ undefined, } BA = \begin{bmatrix} 2 & -9 & -2 \\ 7 & 4 & -1 \end{bmatrix}$$

(b) $Av = [2(1) - 1(-2) + 0(3), 1(1) + 3(-2) - 2(3), 0(1) + 4(-2) + 1(3)]^T = [4, -11, -5]^T$

VA undefined: v is 3×1 so it has 1 col, A has 3 rows, $1 \neq 3$.

$$\text{Ans: } Av = [4, -11, -5]^T, VA \text{ undefined}$$

(c) Aw undefined: A has 3 cols, w is 1×3 so it has 1 row, $3 \neq 1$.

$$wA = [2(2) + 0(1) - 1(0), 2(-1) + 0(3) - 1(4), 2(0) + 0(-2) - 1(1)] = [4, -6, -1]$$

$$\text{Ans: } Aw \text{ undefined, } wA = [4 \ -6 \ -1]$$

(d) vw is $3 \times 1 \cdot 1 \times 3 = 3 \times 3 \rightarrow \begin{bmatrix} 1(2) & 1(0) & 1(-1) \end{bmatrix} \begin{bmatrix} 2 & 0 & -1 \end{bmatrix}$

$$\begin{bmatrix} -2(2) & -2(0) & -2(-1) \end{bmatrix} = \begin{bmatrix} -4 & 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 3(2) & 3(0) & 3(-1) \end{bmatrix} \begin{bmatrix} 6 & 0 & -3 \end{bmatrix}$$

$$wv \text{ is } 1 \times 3 \cdot 3 \times 1 = 1 \times 1 \rightarrow wv = 2(1) + 0(-2) + (-1)(3) = -1$$

$$\text{Ans: } vw \text{ above, } wv = [-1] \ (1 \times 1)$$

(e) $A^2 = AA \rightarrow r1 = [2(2) - 1(1) + 0(0), 2(-1) - 1(3) + 0(4), 2(0) - 1(-2) + 0(1)] = [3, -5, 2]$

$$r2 = [1(2) + 3(1) - 2(0), 1(-1) + 3(3) - 2(4), 1(0) + 3(-2) - 2(1)] = [5, 0, -8]$$

$$r_3 = [0(2) + 4(1) + 1(0), 0(-1) + 4(3) + 1(4), 0(0) + 4(-2) + 1(1)] = [4, 16,$$

-7]

$$\text{Ans: } A^2 = \begin{bmatrix} 3 & -5 & 2 \\ 5 & 0 & -8 \\ 4 & 16 & -7 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 0 & -8 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 16 & -7 \end{bmatrix}$$

(f) B^2 undefined: B is 2×3 , so BB would need 3 cols = 2 rows, which fails. B is not

square, so it cannot be squared.

Ans: B^2 undefined

Problem 3

(a) Augmented matrix $\rightarrow$ eliminate $\rightarrow$ back substitute.

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 2 & 3 & 1 & | & 2 \\ -1 & 1 & 2 & | & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ -1 & 1 & 2 & | & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ -1 & 1 & 2 & | & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ -1 & 1 & 2 & | & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \quad R_3 \rightarrow R_3 + R_1$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ -1 & 1 & 2 & | & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & -1 & 3 & | & 4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 0 & 10 & | & 10 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 3 & 1 & | & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 0 & 10 & | & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & -1 \\ 0 & 1 & -3 & | & -4 \\ 0 & 0 & 1 & | & 1 \end{bmatrix}$$

$$R_2 \rightarrow -R_2 \quad R_3 \rightarrow R_3 - 3R_2 \quad R_3 \rightarrow (1/10)R_3$$

$x_3 = 1 \rightarrow x_2 = -4 + 3(1) = -1 \rightarrow x_1 = -1 - 2(-1) + 1 = 2$. Three pivots, so unique.

$$\text{Ans: } x = [2, -1, 1]^T$$

(b)

$$\begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 2 & 5 & 1 & | & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 0 & 1 & 3 & | & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 2 & 5 & 1 & | & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 0 & 1 & 3 & | & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & -1 & | & 4 \\ 0 & 1 & 3 & | & -2 \end{bmatrix}$$

$$[3 \ 7 \ 0 \mid 10] \quad [3 \ 7 \ 0 \mid 10] \quad [0 \ 1 \ 3 \mid -2]$$

$$R_2 \rightarrow R_2 - 2R_1 \quad R_3 \rightarrow R_3 - 3R_1$$

$$[1 \ 2 \ -1 \mid 4] \quad [1 \ 0 \ -7 \mid 8]$$

$$[0 \ 1 \ 3 \mid -2] \rightarrow [0 \ 1 \ 3 \mid -2]$$

$$[0 \ 0 \ 0 \mid 0] \quad [0 \ 0 \ 0 \mid 0]$$

$$R_3 \rightarrow R_3 - R_2 \quad R_1 \rightarrow R_1 - 2R_2$$

No pivot in col 3 $\rightarrow z$ is free, so there are infinitely many solutions. Let $z = t$:

$$x = 8 + 7t, y = -2 - 3t, z = t.$$

$$\text{Ans: } [x, y, z]^T = [8, -2, 0]^T + t[7, -3, 1]^T, t \in \mathbb{R}$$

Problem 4

(a) Let x_1, x_2, x_3 = mass in kg of Alloy 1, Alloy 2, Alloy 3 used in the blend.

The 100 kg product holds 36 kg Cu, 44 kg Zn, 20 kg Ni, and no metal is lost, so for each metal the mass in = the mass out:

$$\text{Cu: } 0.5x_1 + 0.2x_2 + 0.4x_3 = 36$$

$$\text{Zn: } 0.3x_1 + 0.7x_2 + 0.2x_3 = 44$$

$$\text{Ni: } 0.2x_1 + 0.1x_2 + 0.4x_3 = 20$$

(adding all three $\rightarrow x_1 + x_2 + x_3 = 100$, which checks out)

(b) Multiply each row by 10 to clear the decimals:

$$\text{Ans: } [5 \ 2 \ 4 \mid 360]$$

$$[3 \ 7 \ 2 \mid 440]$$

$$[2 \ 1 \ 4 \mid 200]$$

(c)

$$[5 \ 2 \ 4 \mid 360] \quad [1 \ 0 \ -4 \mid -40] \quad [1 \ 0 \ -4 \mid -40]$$

$$[3 \ 7 \ 2 \mid 440] \rightarrow [3 \ 7 \ 2 \mid 440] \rightarrow [0 \ 7 \ 14 \mid 560]$$

$$\begin{bmatrix} 2 & 1 & 4 & | & 200 \end{bmatrix} \quad \begin{bmatrix} 2 & 1 & 4 & | & 200 \end{bmatrix} \quad \begin{bmatrix} 2 & 1 & 4 & | & 200 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 2R_3 \quad R_2 \rightarrow R_2 - 3R_1$$

$$\begin{bmatrix} 1 & 0 & -4 & | & -40 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & -4 & | & -40 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & -4 & | & -40 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 7 & 14 & | & 560 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 2 & | & 80 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 2 & | & 80 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 12 & | & 280 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 12 & | & 280 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 10 & | & 200 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_1 \quad R_2 \rightarrow (1/7)R_2 \quad R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 0 & -4 & | & -40 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 2 & | & 80 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & | & 20 \end{bmatrix}$$

$$R_3 \rightarrow (1/10)R_3$$

$$x_3 = 20 \rightarrow x_2 = 80 - 2(20) = 40 \rightarrow x_1 = -40 + 4(20) = 40.$$

$$\text{Check: } 40 + 40 + 20 = 100 \text{ kg, Cu} = 20 + 8 + 8 = 36 \text{ kg. } \checkmark$$

$$\text{Ans: } 40 \text{ kg Alloy 1, } 40 \text{ kg Alloy 2, } 20 \text{ kg Alloy 3}$$

Problem 5

(a)

$$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 & 4 \end{bmatrix} \quad \begin{bmatrix} 3 & 6 & 4 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \quad R_3 \rightarrow R_3 - 3R_1 \quad R_3 \rightarrow R_3 - R_2 \quad R_1 \rightarrow R_1 - R_2$$

$$\text{Ans: RREF}(A) = \begin{bmatrix} 1 & 2 & 0; 0 & 0 & 1; 0 & 0 & 0 \end{bmatrix}$$

(b) $\text{rank}(A) = 2$, since the RREF has exactly two pivots (cols 1 and 3), i.e. two nonzero rows.

$$\text{Ans: rank}(A) = 2$$

(c) Row reducing $[A \mid b]$ the same way makes row 3 into $[0 \ 0 \ 0 \mid b_3 - b_1 - b_2]$.

0 solutions: possible. If $b_3 \neq b_1 + b_2$, e.g. $b = [0, 0, 1]^T$, row 3 reads $0 = 1$, so the system is inconsistent.

Exactly 1: impossible. Col 2 is never a pivot col, so x_2 is free whenever the system is consistent; $\text{rank}(A) = 2 < 3$ unknowns, so uniqueness cannot happen.

Infinitely many: possible. If $b_3 = b_1 + b_2$, e.g. $b = 0$ or $b = [1, 3, 4]^T$, the system is consistent and the free variable x_2 gives a whole line of solutions.

Ans: 0 possible, 1 impossible, ∞ possible

Problem 6

(a) Row reduce the matrix whose columns are v_1, v_2, v_3 :

$$\begin{array}{cccc} [1 & 2 & 0] & [1 & 2 & 0] & [1 & 2 & 0] & [1 & 0 & -2] \\ [0 & 1 & 1] \rightarrow [0 & 1 & 1] \rightarrow [0 & 1 & 1] \rightarrow [0 & 1 & 1] \\ [2 & 3 & -1] & [0 & -1 & -1] & [0 & 0 & 0] & [0 & 0 & 0] \end{array}$$

$$R_3 \rightarrow R_3 - 2R_1 \quad R_3 \rightarrow R_3 + R_2 \quad R_1 \rightarrow R_1 - 2R_2$$

Two pivots for three vectors $\rightarrow$ linearly dependent. The relation is $v_3 = v_2 - 2v_1$, since $[2, 1, 3]^T - [2, 0, 4]^T = [0, 1, -1]^T$. ✓

v_1 and v_2 are not parallel, so their span is a plane through the origin.

Ans: dependent; span is a plane in $\mathbb{R}^3$

(b) Same idea with v_1, v_2, v_3 as columns:

$$\begin{array}{ccc} [1 & 0 & 1] & [1 & 0 & 1] & [1 & 0 & 1] \\ [1 & 1 & 0] \rightarrow [0 & 1 & -1] \rightarrow [0 & 1 & -1] \end{array}$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1 \quad R_3 \rightarrow R_3 - R_2$$

Three pivots for three vectors $\rightarrow c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$ only when $c_1 = c_2 = c_3 = 0$, so the set is linearly independent.

Three independent vectors in $\mathbb{R}^3$ span all of $\mathbb{R}^3$.

Ans: independent; span is all of $\mathbb{R}^3$
